CONDITIONAL HETEROSKEDASTICITY DRIVEN BY HIDDEN MARKOV CHAINS

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Abstract. We consider a generalized autoregressive conditionally heteroskedastic (GARCH) equation where the coefficients depend on the state of a nonobserved Markov chain. Necessary and sufficient conditions ensuring the existence of a stationary solution are given. In the case of ARCH regimes, the maximum likelihood estimates are shown to be consistent. The identification problem is also considered. This is illustrated by means of real and simulated data sets.

Keywords. Nonlinear time series models; ARCH models; hidden Markov chain; switching models; stationary solution; maximum likelihood; consistency.

1. INTRODUCTION

Since the seminal works of Engle (1982) and Bollerslev (1986), generalized autoregressive conditionally heteroskedastic (GARCH) models have been the most widely used to model time-varying conditional variances. In these models the conditional variance changes over time as a fixed function of the past. However, it has been shown empirically that some financial series typically exhibit changes in the dynamics of the conditional variance (see for instance Hamilton and Susmel, 1994). One popular approach to modeling changes in regimes is the class of Markov-switching models introduced by Hamilton (1989) in the econometric literature.[1] For this reason, Hamilton and Susmel (1994), Cai (1994) and Dueker (1997) considered Markov-switching parameters to GARCH models for modeling financial time series. In this paper we consider a GARCH equation where the coefficients depend on the state of a nonobserved Markov chain and we concentrate on two problems that were not completely treated by the above-mentioned authors. First, we give necessary and sufficient conditions ensuring the existence of a stationary solution. We find that the GARCH coefficients associated with a given state of the Markov chain can violate the stationarity conditions of standard GARCH models. In other words we show that explosive GARCH parameters are possible for subsamples of a stationary time series (in practice GARCH parameter estimates close to the nonstationary region are often observed). Second, the asymptotic properties of the maximum likelihood estimator are investigated. Consistency is shown in the case of

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ARCH regimes and it is demonstrated that information criteria such as the Akaike information criterion (AIC) or the Bayesian information criterion (BIC) do not underparameterize the model for large samples. This is illustrated by means of real and simulated data sets. In particular we show that, for a series with at least two GARCH regimes, the standard GARCH parameter estimates are generally explosive. This is a plausible explanation for the fact that most of the applications of GARCH models lead to parameter estimates close to the nonstationary region. Thus explosive GARCH parameter estimates should incline the practitioner to try models with volatility changes over time, like the model proposed in this paper.

The paper is organized as follows. The model to be considered is described in Section 2. Stationarity conditions are given in Section 3. The maximum likelihood procedure is presented and its consistency is proved in Section 4. Section 5 is devoted to identification problems. Numerical illustrations are presented in Section 6.

2. GARCH MODELS DRIVEN BY HIDDEN MARKOV CHAINS

Consider a process $(\epsilon_t)_{t\in\mathbb{Z}}$ satisfying an equation of the form

$$\begin{aligned} \epsilon_t &= (h_t)^{1/2}\eta_t \\ h_t &= \omega(\Delta_t) + \sum_{i=1}^{q}\alpha_i(\Delta_t)\epsilon_{t-i}^2 + \sum_{j=1}^{p}\beta_j(\Delta_t)h_{t-j} \end{aligned} \tag{1}$$

where (η_t) is a sequence of independent and identically distributed (i.i.d.) random variables with zero mean and unit variance, and (Δ_t) is a Markov chain with finite state-space $\mathscr{E} = \{1, 2, \ldots, d\}$. Given $\{\Delta_t = k\}$, $k \in \mathscr{E}$, ϵ_t satisfies a GARCH equation with non-negative coefficients $\alpha_i(k)$, $1 \leqslant i \leqslant q$, $\beta_j(k)$, $1 \leqslant j \leqslant p$, and a positive coefficient $\omega(k)$. These standard assumptions on the GARCH coefficients entail that h_t is almost surely strictly positive. Neither the noise (η_t) nor the Markov chain (Δ_t) are observed (the Markov chain (Δ_t) is said to be hidden). Suppose that (η_t) and (Δ_t) are independent and that (Δ_t) is stationary, irreducible and aperiodic. Let $\pi(i) := P(\Delta_1 = i)$, $1 \leqslant i \leqslant d$, be the stationary distribution and $p(i, j) := P(\Delta_t = j/\Delta_{t-1} = i)$ be the transition probabilities. Note that the assumptions made on the Markov chain imply that $\pi(i) \neq 0$, $\forall i \in \mathscr{E}$.

3. NECESSARY AND SUFFICIENT STATIONARITY CONDITIONS

The aim of this section is to study the existence of strictly stationary and second-order stationary solutions of (1). The second equation in (1) can be rewritten in vectorial form as

$$\underline{h}_t = \underline{\omega}_t + A_t\underline{h}_{t-1} \tag{2}$$

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where $r = \max(p, q)$,

$$\underline{\omega}_t = \underline{\omega}(\Delta_t) = \begin{pmatrix} \omega(\Delta_t) \\ 0 \\ \vdots \\ 0 \end{pmatrix} \in \mathbb{R}^r \qquad \underline{h}_t = \begin{pmatrix} h_t \\ h_{t-1} \\ \vdots \\ h_{t-r+1} \end{pmatrix} \in \mathbb{R}^r$$

and

$$A_t = \begin{pmatrix} \alpha_1(\Delta_t)\eta_{t-1}^2 + \beta_1(\Delta_t) & \alpha_2(\Delta_t)\eta_{t-2}^2 + \beta_2(\Delta_t) & \cdots & \alpha_r(\Delta_t)\eta_{t-r}^2 + \beta_r(\Delta_t) \\ 1 & 0 & \cdots & 0 \\ 0 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 \quad 1 & 0 \end{pmatrix}$$

is an $r \times r$ matrix (by convention the $\alpha_i(\Delta_t)$, $\beta_j(\Delta_t)$ are equal to zero for $i > q$ and $j > p$). Note that the existence of a nonanticipative (i.e. measurable with respect to the σ-field generated by $(\Delta_u, \eta_u, u \leqslant t)$) stationary solution of (1) is equivalent to the existence of a nonanticipative stationary solution of (2). Since (Δ_t, η_t) is an ergodic process, $(A_t, \underline{\omega}_t)$ is also a strictly stationary ergodic process. It is clear that both $E\log^+\|A_t\|$ and $E\log^+\|\underline{\omega}_t\|$ are finite. Therefore form Brandt (1986) (see Bougerol and Picard, 1992a) the unique stationary solution of (2) is given by

$$\underline{h}_t = \underline{\omega}_t + \sum_{k=1}^{\infty} A_t A_{t-1} \cdots A_{t-k+1} \underline{\omega}_{t-k} \tag{3}$$

whenever the top Lyapunov exponent γ defined by

$$\gamma = \inf\left\{ E\frac{1}{n}\log\|A_n A_{n-1} \cdots A_1\|,\ n \in \mathbb{N}^* \right\}$$

is strictly negative. Since all the norms are equivalent on a finite-dimensional vectorial space, the choice of the norm is unimportant in this definition. The following theorem states that the converse of Brandt's result is also true in our context.

THEOREM 1. *Equation (2) has a strictly stationary solution if and only if the top Lyapunov exponent γ associated with $(A_t)_{t\in\mathbb{Z}}$ is strictly negative. Moreover, when $\gamma < 0$, the stationary solution is unique, ergodic, nonanticipative and defined by (3).*

PROOF. We have already mentioned that the 'if' part follows from Brandt (1986) or Bougerol and Picard (1992a, Theorem 1.1). For the converse, it suffices to show (see Bougerol and Picard, 1992a, Lemma 3.4) that

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$$\lim_{n\to\infty} \|A_0 A_{-1} \cdots A_{-n}\| = 0 \qquad \text{almost surely (a.s.).} \tag{4}$$

Following Bougerol and Picard (1992b, proof of Theorem 1.3) we will show that, for $1 \leqslant i \leqslant r$,

$$\lim_{n\to\infty} A_0 A_{-1} \cdots A_{-n} \boldsymbol{e}_i = 0 \tag{5}$$

where $\boldsymbol{e}_i$ is the ith element of the canonical basis of $\mathbb{R}^r$. First note that if $(\underline{h}_t)$ is a strictly stationary solution of (2) then

$$\underline{h}_0 = \underline{\omega}_0 + \sum_{k=0}^{n} A_0 A_{-1} \cdots A_{-k} \underline{\omega}_{-k-1} + A_0 A_{-1} \cdots A_{-n-1} \underline{h}_{-n-2}$$

$$\geqslant \sum_{k=0}^{n} A_0 A_{-1} \cdots A_{-k} \underline{\omega}_{-k-1}$$

since all the involved matrices are non-negative. Therefore the series $\sum_{k=0}^{n} A_0 A_{-1} \cdots A_{-k} \underline{\omega}_{-k-1}$ converges almost surely and $A_0 A_{-1} \cdots A_{-n} \underline{\omega}_{-n-1} \to 0$ a.s. as $n \to \infty$. Since $\underline{\omega}_{-n-1} \geqslant \min_{j\in\mathscr{E}} \omega(j) \boldsymbol{e}_1$, (5) holds for $i = 1$. For $1 < k \leqslant r$, we have

$$A_{-n+k-2} \cdots A_{-n} \underline{\omega}_{-n-1} \geqslant \min_{j\in\mathscr{E}} \omega(j) \boldsymbol{e}_k.$$

Therefore (5) holds for all the is, which completes the proof. ■

In the case of the standard GARCH(1, 1) model (i.e. $d = q = p = 1$), the previous theorem reduces to the result given by Nelson (1990), since $\gamma = E \log\{\alpha_1(1)\eta_t^2 + \beta_1(1)\}$. More generally, keeping $r = \max(p, q) = 1$ and for an arbitrary number d of regimes, the coefficient becomes

$$\gamma = \sum_{k=1}^{d} \pi(k) E \log\{\alpha_1(k)\eta_t^2 + \beta_1(k)\}.$$

It is interesting to note that it is not necessary that the Nelson stationarity condition be satisfied for all the GARCH regimes but only on average with respect to the stationary distribution of the chain. In the general case, the Lyapunov exponent γ seems difficult to compute but it is easily estimated by Monte Carlo simulations since

$$\gamma = \lim_{n\to\infty} \frac{1}{n} \log\|A_n A_{n-1} \cdots A_1\| \text{ a.s.}$$

It should be noted that the condition $\gamma < 0$ does not entail the existence of an unconditional variance for (ϵ_t). The following theorem provides necessary and sufficient conditions for the existence of a second-order stationarity solution of (1).

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THEOREM 2. *Let $\tilde{A}(\Delta_t)$ be the $r \times r$ matrix obtained by replacing η_{t-i}^2 by 1 in A_t, for $i = 1, \ldots, r$. Let the $dr \times dr$ matrix*

$$\boldsymbol{P} = \begin{pmatrix} p(1,1)\tilde{A}(1) & p(2,1)\tilde{A}(1) & \cdots & p(d,1)\tilde{A}(1) \\ p(1,2)\tilde{A}(2) & p(2,2)\tilde{A}(2) & \cdots & p(d,2)\tilde{A}(2) \\ \vdots & & & \\ p(1,d)\tilde{A}(d) & p(2,d)\tilde{A}(d) & \cdots & p(d,d)\tilde{A}(d) \end{pmatrix}.$$

If the spectral radius of $\boldsymbol{P}$, denoted by $\rho(\boldsymbol{P})$, is strictly less than 1 then (2) admits one and only one strictly stationary solution. This solution belongs to L^1; it is the ergodic process defined by (3). The variance of $\epsilon_t = (h_t)^{1/2}\eta_t$ is then given by

$$\operatorname{var}(\epsilon_t) = Eh_t = \frac{1}{r}\sum_{k=0}^{\infty} \mathbf{1}'\boldsymbol{P}^k\boldsymbol{\omega}$$

where

$$\mathbf{1}' = (1, \ldots, 1) \qquad \boldsymbol{\omega} = (\pi(1)\underline{\omega}'(1), \ldots, \pi(d)\underline{\omega}'(d))'$$

are respectively $1 \times dr$ and $dr \times 1$ matrices.

Conversely, suppose that $\boldsymbol{P}$ is irreducible and that $\rho(\boldsymbol{P}) \geqslant 1$; then there exists no strictly stationary solution of (2) belonging to L^1.

PROOF. First define, for $t, k \in \mathbb{Z}$, the following $\mathbb{R}^r$-valued random vectors:

$$H_k(t) = \begin{cases} 0 & \text{if } k < 0 \\ \underline{\omega}_t + A_t H_{k-1}(t-1) & \text{if } k \geqslant 0. \end{cases}$$

We have

$$H_k(t) - H_{k-1}(t) = \begin{cases} 0 & \text{if } k < 0 \\ \underline{\omega}_t & \text{if } k = 0 \\ A_t\{H_{k-1}(t-1) - H_{k-2}(t-1)\} & \text{if } k > 0. \end{cases}$$

Throughout, we consider the matrix norm defined by $\|A\| = \sum_{i,j}|A(i,j)|$ where $A(i,j)$ denotes the generic element of A. Since $A_tA_{t-1}\cdots A_{t-k+1}\underline{\omega}_{t-k}$ has positive elements, for $k > 0$ we have

$$\begin{aligned} E\|H_k(t) - H_{k-1}(t)\| &= E\|A_tA_{t-1}\cdots A_{t-k+1}\underline{\omega}_{t-k}\| \\ &= \|EA_t\cdots A_{t-k+1}\underline{\omega}_{t-k}\| \\ &= \|E\{E(A_t\cdots A_{t-k+1}\underline{\omega}_{t-k}|\Delta_t, \Delta_{t-1}, \ldots, \Delta_{t-k})\}\| \\ &= \|E\{\tilde{A}(\Delta_t)\cdots\tilde{A}(\Delta_{t-k+1})\underline{\omega}_{t-k}\}\| \\ &= \|\mathbb{I}\boldsymbol{P}^k\boldsymbol{\omega}\| = \mathbf{1}'\boldsymbol{P}^k\boldsymbol{\omega} \end{aligned}$$

where $\mathbb{I}' = (I_r, \ldots, I_r)$ and I_r are respectively an $r \times dr$ matrix and the $r \times r$

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identity matrix. When $\rho(\boldsymbol{P})<1$, using the Jordan decomposition of $\boldsymbol{P}$, it is easy to show that $\mathbf{1}'\boldsymbol{P}^k\boldsymbol{\omega}$ tends to zero at an exponential rate as $k\to\infty$. Hence, for t fixed, $H_k(t)$ converges both in L^1 and almost surely as $k\to\infty$. Write $\underline{h}_t$ as the limit in L^1 and almost sure of $\{H_k(t)\}_k$. Since, for all k, $\{H_k(t)\}_{t\in\mathbb{Z}}$ is strictly stationary, the limit $(\underline{h}_t)_{t\in\mathbb{Z}}$ is also strictly stationary. It is obvious that $(\underline{h}_t)_{t\in\mathbb{Z}}$ satisfies (2) and (3). The uniqueness of the stationary solution is proved without further difficulties.

Now suppose that $(\underline{h}_t)$ is a solution of (2) belonging to L^1. We have

$$
\begin{aligned}
E\underline{h}_t &= E\underline{\omega}_t + EA_t\underline{\omega}_{t-1} + \cdots + EA_t\cdots A_{t-k+1}\underline{h}_{t-k}\\
&\geqslant E\underline{\omega}_t + \sum_{k=1}^{\infty} EA_t\cdots A_{t-k+1}\underline{\omega}_{t-k}\\
&= \sum_{k=0}^{\infty} \mathbb{I}'\boldsymbol{P}^k\boldsymbol{\omega}\\
&\geqslant \frac{1}{(dr)!}\sum_{k=0}^{\infty} \mathbb{I}'\boldsymbol{P}^{k\,dr}(I_{dr}+\boldsymbol{P})^{dr-1}\boldsymbol{\omega}
\end{aligned}
$$

since all the involved random variables are positive. Since $\boldsymbol{P}$ is non-negative and irreducible, $(I_{dr}+\boldsymbol{P})^{dr-1}>0$ (see, for example, Lancaster and Tismenetsky, 1985, p. 533). Therefore, $(I_{dr}+\boldsymbol{P})^{dr-1}\boldsymbol{\omega}>0$ and

$$
\|\mathbb{I}'\boldsymbol{P}^{k\,dr}(I_{dr}+\boldsymbol{P})^{dr-1}\boldsymbol{\omega}\| \geqslant c\|\boldsymbol{P}^{k\,dr}\| \geqslant c\rho(\boldsymbol{P})^{k\,dr}
$$

where c is the strictly positive constant equal to the smallest element of $(I_{dr}+\boldsymbol{P})^{dr-1}\boldsymbol{\omega}$ and $\|\cdot\|$ denotes the matrix norm defined as the sum of the absolute values of all the elements. Therefore, for $E\epsilon_t^2$ to be finite it is necessary that $\rho(\boldsymbol{P})<1$. ■

REMARK 1. In the case $d=1$, which corresponds to a standard GARCH model, matrix $\boldsymbol{P}=\tilde{A}(1)$ is irreducible when r is such that $\alpha_r(1)+\beta_r(1)\neq 0$ (which can be supposed without loss of generality). It is easy to show that the nonzero eigenvalues of $\tilde{A}(1)$ are the zeros of the polynomial $\mathscr{P}(z)=z^r-\sum_{i=1}^{r}\{\alpha_i(1)+\beta_i(1)\}z^{r-i}$ and that this polynomial has its zeros inside the open unit disk if and only if

$$
\sum_{i=0}^{r}\alpha_i(1)+\beta_i(1)<1 \tag{6}
$$

which is the condition given by Bollerslev (1986).

REMARK 2. Cai (1994) considered a particular case of two ARCH regimes with a variable intercept $\omega(\Delta_t)$ and regime-constant values for the other ARCH parameters ($\alpha_i(\Delta_t)\equiv\alpha_i$). In this case $\tilde{A}(\Delta_t)\equiv\tilde{A}$ and $\boldsymbol{P}=\{p(i,j)\}\otimes\tilde{A}$, where

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$\otimes$ denotes the Kronecker product of matrices. Therefore the eigenvalues of $\boldsymbol{P}$ are the products of the eigenvalues of $\{p(i, j)\}$ and $\tilde{A}$. Since the spectral radius of $\{p(i, j)\}$ is 1, the spectral radius of $\boldsymbol{P}$ is equal to that of $\tilde{A}$. From Remark 1 we have $\rho(\tilde{A}) < 1$ if and only if $\sum_i \alpha_i < 1$, which is the condition given by Cai (1994). Therefore we have shown that, under the assumption given on (Δ_t) and $\boldsymbol{P}$, the sufficient condition given by Cai (1994) is also necessary. More generally, for GARCH models when only the intercept $\omega(\Delta_t)$ is switching, the same arguments show that the stationary condition is still the Bollerslev condition (6) of standard GARCH models.

4. MAXIMUM LIKELIHOOD ESTIMATOR

In this section the orders p, q and d are supposed to be known. The maximum likelihood approach requires specification of a particular distribution. Therefore suppose that η_t is normally distributed with mean zero with variance 1. Note, however, that the results given below can be extended to many other distributions. Let $\omega = (\omega(1), \ldots, \omega(d))' \in \mathbb{R}^d$, $\alpha(k) = (\alpha_1(k), \ldots, \alpha_q(k))' \in \mathbb{R}^q$ and $\beta(k) = (\beta_1(k), \ldots, \beta_p(k))' \in \mathbb{R}^p (k = 1, \ldots, d)$ be the coefficients of the GARCH models. Let θ be a vector containing these GARCH parameters and the transition probabilities. The definition will be made more formal below. Let $(\epsilon_1, \ldots, \epsilon_n)$ be a realization of length n of the nonanticipative stationary solution of (1). Given initial values that will be specified below, the conditional likelihood $L_\theta(\epsilon_1, \ldots, \epsilon_n)$ is given by summing, over all the possible paths of the Markov chain, the probability density at the point $(\epsilon_1, \ldots, \epsilon_n)$ given a particular path:

$$L_\theta(\epsilon_1, \ldots, \epsilon_n) = \sum_{(e_1,\ldots,e_n)\in \mathscr{E}^n} \pi(e_1)\left\{\prod_{t=2}^{n} p(e_{t-1}, e_t)\right\}\left\{\prod_{t=1}^{n} f_{e_1,\ldots,e_t}(\epsilon_1, \ldots, \epsilon_t)\right\} \quad (7)$$

where

$$f_{e_1,\ldots,e_t}(\epsilon_1, \ldots, \epsilon_t) = \frac{1}{(2\pi)^{1/2}\sigma_{e_1,\ldots,e_t}(\epsilon_1, \ldots, \epsilon_{t-1})} \exp\left\{-\frac{\epsilon_t^2}{2\sigma^2_{e_1,\ldots,e_t}(\epsilon_1, \ldots, \epsilon_{t-1})}\right\}$$

and the $\sigma_{e_1,\ldots,e_t}(\epsilon_1, \ldots, \epsilon_{t-1})$ are defined recursively by

$$\sigma^2_{e_1,\ldots,e_t}(\epsilon_1, \ldots, \epsilon_{t-1}) = \omega(e_t) + \sum_{i=1}^{q} \alpha_i(e_t)\epsilon^2_{t-i} + \sum_{j=1}^{p} \beta_j(e_t)\sigma^2_{e_1,\ldots,e_{t-j}}(\epsilon_1, \ldots, \epsilon_{t-j-1})$$

for $t = p + 1, \ldots, n$, and

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$$\sigma^2_{e_1,\ldots,e_t}(\epsilon_1, \ldots, \epsilon_{t-1}) = \omega(e_t) + \sum_{i=1}^{q} \alpha_i(e_t)\epsilon^2_{t-i} + \sum_{j=1}^{t-1} \beta_j(e_t)\sigma^2_{e_1,\ldots,e_{t-j}}(\epsilon_1, \ldots, \epsilon_{t-j-1}) + \sum_{j=t}^{p} \beta_j(e_t)\tilde{\sigma}^2_{t-j}$$

for $t = 1, \ldots, p$ (for $t = 1$, $\sigma^2_{e_1,\ldots,e_t}(\epsilon_1, \ldots, \epsilon_{t-1})$ denotes a constant depending only on e_1, and by convention $\sum_{j=1}^{t-1} = 0$ for $t \leqslant 1$ and $\sum_{j=t}^{p} = 0$ for $t \geqslant p$), and where

$$\epsilon_0, \ldots, \epsilon_{1-q}, \tilde{\sigma}^2_0, \ldots, \tilde{\sigma}^2_{1-p}$$

are given initial values. The direct computation of the likelihood becomes quickly intractable as n increases. In the case $p = 0$, $\sigma_{e_1,\ldots,e_t}(\epsilon_1, \ldots, \epsilon_{t-1})$ depends only on e_t and on the last q values $\epsilon_{t-q}, \ldots, \epsilon_{t-1}$. Using this Markov property and writing $f_{e_1,\ldots,e_t}(\epsilon_1, \ldots, \epsilon_t) = f_{e_t}(\epsilon_{t-q}, \ldots, \epsilon_t)$ the likelihood can be rewritten as a product of matrices.

Let $\mathbf{1} = (1, \ldots, 1)' \in \mathbb{R}^d$, $\boldsymbol{p}(\epsilon_1) = (\pi(1)f_1(\epsilon_{1-q}, \ldots, \epsilon_1), \ldots, \pi(d)\ f_d(\epsilon_{1-q}, \ldots, \epsilon_1))' \in \mathbb{R}^d$ and

$$M_\theta(\epsilon_{t-q}, \ldots, \epsilon_t) = \begin{pmatrix} p(1,1)f_1(\epsilon_{t-q}, \ldots, \epsilon_t) & p(2,1)f_1(\epsilon_{t-q}, \ldots, \epsilon_t) & \cdots & p(d,1)f_1(\epsilon_{t-q}, \ldots, \epsilon_t) \\ p(1,2)f_2(\epsilon_{t-q}, \ldots, \epsilon_t) & p(2,2)f_2(\epsilon_{t-q}, \ldots, \epsilon_t) & \cdots & p(d,2)f_2(\epsilon_{t-q}, \ldots, \epsilon_t) \\ \vdots & \vdots & & \vdots \\ p(1,d)f_d(\epsilon_{t-q}, \ldots, \epsilon_t) & p(2,d)f_d(\epsilon_{t-q}, \ldots, \epsilon_t) & \cdots & p(d,d)f_d(\epsilon_{t-q}, \ldots, \epsilon_t) \end{pmatrix}.$$

Then it is easy to check that, when $p = 0$,

$$L_\theta(\epsilon_1, \ldots, \epsilon_n) = \mathbf{1}' \left\{ \prod_{t=2}^{n} M_\theta(\epsilon_{n+2-t-q}, \ldots, \epsilon_{n-t+2}) \right\} \boldsymbol{p}(\epsilon_1). \tag{8}$$

From (8) it is easy to compute $L_\theta(\epsilon_1, \ldots, \epsilon_n)$ recursively: for $k = 2, \ldots, n$

$$\left\{ \prod_{t=2}^{k} M_\theta(\epsilon_{k+2-t-q}, \ldots, \epsilon_{k+2-t}) \right\} = M_\theta(\epsilon_{k-q}, \ldots, \epsilon_k) \left\{ \prod_{t=2}^{k-1} M_\theta(\epsilon_{k+1-t-q}, \ldots, \epsilon_{k+1-t}) \right\}. \tag{9}$$

In fact, (9) is (the backward part of) a recursive algorithm usually called the forward–backward algorithm (see Baum, 1972). Note that the procedure proposed by Hamilton and Susmel (1994) can also be easily adapted to our model, in order to compute the log-likelihood function recursively. This procedure has been used for the implementation of the numerical illustrations presented below. When $p \neq 0$, the complexity of the likelihood function is such

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that more advanced techniques, like the simulation techniques proposed by Shephard (1994), and Billio and Monfort (1995), should be used to derive an inference on θ. However, by using these simulation techniques, one loses the computational simplicity of the forward–backward algorithm (9). Moreover it seems to be a real challenge to state the consistency of the underlying estimators. For these reasons we focus on the maximum likelihood estimator in the case $p = 0$. We will prove the consistency of this estimator.

By definition the transition probabilities are subject to the constraints $\sum_{j=1}^{d} p(i, j) = 1$, $\forall i \in \mathscr{E}$. So the unknown parameter can be chosen to be of the form

$$\theta := (p(1, 2),\ p(1, 3),\ \ldots,\ p(1, d),\ p(2, 2),\ \ldots,\ p(d, d),\ \omega',\ \alpha(1)',\ \ldots,\ \alpha(d)')'.$$

The parameter space Θ is defined as the subset of $\mathbb{R}^{d^2+qd}$ of the parameters compatible with the assumptions made in Sections 2 and 3, i.e. for all $\theta \in \Theta$, the associated Markov chain is irreducible and aperiodic and the GARCH parameters satisfy $\rho(\boldsymbol{P}) < 1$. It is supposed that the true parameter value θ_0 belongs to Θ. We denote by $g_{m,\theta}(\cdot|\epsilon_{t-m}, \epsilon_{t-m-1}, \ldots)$ the density function of $(\epsilon_t, \ldots, \epsilon_{t-m+1})$ given the σ-field generated by $\epsilon_{t-m}, \epsilon_{t-m-1}, \ldots$. In addition to the previous assumptions, the following identifiability condition is imposed. As will be shown in the example below, it entails restrictions on the choice of the parameter space Θ.

CONDITION A. For all $\theta \in \Theta$, if $g_{1,\theta}(\epsilon_t|\epsilon_{t-1}, \epsilon_{t-2}, \ldots) = g_{1,\theta_0}(\epsilon_t|\epsilon_{t-1}, \epsilon_{t-2}, \ldots)$ P_{θ_0} − a.s, then $\theta = \theta_0$.

As we shall see in Example 1 below, it may be convenient to check Condition A by showing that there exists a positive integer m such that, for $\theta \neq \theta_0$, the random variables $g_{m,\theta}(\epsilon_t, \ldots, \epsilon_{t-m+1}|\epsilon_{t-m}, \epsilon_{t-m-1}, \ldots)$ and $g_{m,\theta_0}(\epsilon_t, \ldots, \epsilon_{t-m+1}|\epsilon_{t-m}, \epsilon_{t-m-1}, \ldots)$ are not equal P_{θ_0} − almost surely.

EXAMPLE 1. Consider the case $d = 2$, $p = 0$ and $q = 1$. Let

$$\begin{aligned}\Theta = \{&\theta = (p(1, 2),\ p(2, 1),\ \omega(1),\ \omega(2),\ \alpha(1),\ \alpha(2))' \\ &\in]0, 1[^2 \times]0, +\infty[^2 \times]0, +\infty[^2: \\ &p(1, 2) < p(2, 1),\ [1 - p(1, 2)]\alpha(1) + p(1, 2)\alpha(2) \leqslant 1, \\ &p(2, 1)\alpha(1) + [1 - p(2, 1)]\alpha(2) < 1,\ \omega(1) \neq \omega(2)\}.\end{aligned}$$

Then $\rho(\boldsymbol{P}) < 1$ for all $\theta \in \Theta$, and Condition A holds.

PROOF. The Geršgorin theorem (see, for example, Lancaster and Tismenetsky, 1985, p. 377) shows that the stationary condition $\rho(\boldsymbol{P}) < 1$ is satisfied. Let $\theta = (p(1, 2),\ p(2, 1),\ \omega(1),\ \omega(2),\ \alpha(1),\ \alpha(2))'$ and $\theta_0 = (p_0(1, 2),\ p_0(2, 1),\ \omega_0(1),\ \omega_0(2),\ \alpha_0(1),\ \alpha_0(2))'$ be two elements of Θ such that

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$$g_{1,\theta}(\epsilon_t|\epsilon_{t-1}, \epsilon_{t-2}, \ldots) = g_{1,\theta_0}(\epsilon_t|\epsilon_{t-1}, \epsilon_{t-2}, \ldots) \qquad P_{\theta_0} - \text{a.s.}$$

Now, given $\epsilon_{t-1}, \epsilon_{t-2}, \ldots$, the conditional distribution of ϵ_t is a mixture of two Gaussian distributions with zero means, respective covariances

$$\omega(i) + \alpha(i)\epsilon_{t-1}^2$$

and respective weights

$$P_\theta(\Delta_t = i|\epsilon_{t-1}, \epsilon_{t-2}, \ldots) \qquad \text{for } i = 1, 2.$$

From Teicher (1963) the parameters and the weights of these two Gaussian distributions can be determined from the mixing distribution. In other words we have almost surely

$$\sum_{i=1}^{2}\{\omega(i) + \alpha(i)\epsilon_{t-1}^2\}\delta_{P_\theta(\Delta_t=i|\epsilon_{t-1},\epsilon_{t-2},\ldots)}$$

$$= \sum_{i=1}^{2}\{\omega_0(i) + \alpha_0(i)\epsilon_{t-1}^2\}\delta_{P_{\theta_0}(\Delta_t=i|\epsilon_{t-1},\epsilon_{t-2},\ldots)} \tag{10}$$

where δ_x denotes the point mass at x. Then we have either case 1,

$$\omega(1) = \omega_0(1), \omega(2) = \omega_0(2), \alpha(1) = \alpha_0(1), \alpha(2) = \alpha_0(2)$$

$$P_\theta(\Delta_t = i|\epsilon_{t-1}, \epsilon_{t-2}, \ldots) = P_{\theta_0}(\Delta_t = i|\epsilon_{t-1}, \epsilon_{t-2}, \ldots) \qquad i = 1, 2$$

or case 2,

$$\omega(1) = \omega_0(2), \omega(2) = \omega_0(1), \alpha(1) = \alpha_0(2), \alpha(2) = \alpha_0(1)$$

$$P_\theta(\Delta_t = 1|\epsilon_{t-1}, \epsilon_{t-2}, \ldots) = P_{\theta_0}(\Delta_t = 2|\epsilon_{t-1}, \epsilon_{t-2}, \ldots)$$

$$P_\theta(\Delta_t = 2|\epsilon_{t-1}, \epsilon_{t-2}, \ldots) = P_{\theta_0}(\Delta_t = 1|\epsilon_{t-1}, \epsilon_{t-2}, \ldots).$$

By stationarity we have

$$g_{2,\theta}(\epsilon_t, \epsilon_{t-1}|\epsilon_{t-2}, \epsilon_{t-3}, \ldots) = g_{2,\theta_0}(\epsilon_t, \epsilon_{t-1}|\epsilon_{t-2}, \epsilon_{t-3}, \ldots) \qquad P_{\theta_0} - \text{a.s.}$$

The marginals of the conditional laws are therefore equal. Let $\tilde{g}_\theta(\cdot|\epsilon_{t-2}, \epsilon_{t-3}, \ldots)$ be the density of the first one. We have

$$\tilde{g}_\theta(\epsilon_t|\epsilon_{t-2}, \epsilon_{t-3}, \ldots) = \tilde{g}_{\theta_0}(\epsilon_t|\epsilon_{t-2}, \epsilon_{t-3}, \ldots) \qquad P_{\theta_0} - \text{a.s.}$$

The conditional distribution of ϵ_t given $\epsilon_{t-2}, \epsilon_{t-3}, \ldots$ is a mixture of the four distributions of the variables $Z(i, j)$, $i, j = 1, 2$, given by

$$Z(i, j) = [\omega(j) + \alpha(j)\eta_{t-1}^2\{\omega(i) + \alpha(i)\epsilon_{t-2}^2\}]^{1/2}\eta_t$$

with respective weights

$$p_\theta(i, j)P_\theta(\Delta_{t-1} = i|\epsilon_{t-2}, \epsilon_{t-3}, \ldots).$$

In particular

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$$E_\theta(\epsilon_t^{2k}|\epsilon_{t-2}, \epsilon_{t-3}, \ldots)$$

$$= \sum_{i,j=1}^{2} p_\theta(i, j)P_\theta(\Delta_{t-1} = i|\epsilon_{t-2}, \epsilon_{t-3}, \ldots)E_\theta\{Z^{2k}(i, j)|\epsilon_{t-2}, \epsilon_{t-3}, \ldots\}$$

for all $k = 1, 2, \ldots$. By equating the second and fourth moments ($k = 1$ and 2) and the conditional distribution of ϵ_t given $\epsilon_{t-2}, \epsilon_{t-3}, \ldots$ for θ and θ_0, some tedious computations show that we have necessarily

$$p_\theta(i, j) = p_{\theta_0}(i, j) \qquad \text{for } i, j = 1, 2$$

in case 1 and

$$p_\theta(i, j) = p_{\theta_0}(j, i) \qquad \text{for } i, j = 1, 2$$

in case 2. The latter situation is precluded since $p_\theta(1, 2) < p_\theta(2, 1)$ for all $\theta \in \Theta$. Therefore $\theta = \theta_0$, which shows that Condition A is satisfied. ■

The proof of the consistency of the maximum likelihood estimator relies on a set of intermediate results which we will present in the following lemmas.

LEMMA 1. *For all $i \in \mathscr{E}$ and $\theta \in \Theta$ we have*

$$P_\theta(\Delta_t = i|\epsilon_{t-1}, \epsilon_{t-2}, \ldots) > 0 \qquad P_{\theta_0} - \text{a.s.}$$

PROOF. The irreductibility and aperiodicity assumptions imply that (Δ_t) is primitive (see Seneta, 1981, p. 21), i.e. there exists a strictly positive integer k such that

$$\forall (i, j) \in \mathscr{E}^2 \qquad P_\theta(\Delta_t = j|\Delta_{t-k} = i) > 0.$$

Let $g_\theta(\cdot|\Delta_t = i, \Delta_{t-k} = j)$ and $g_\theta(\cdot|\Delta_{t-k} = j)$ be the density of $(\epsilon_{t-1}, \ldots, \epsilon_{t-k-q+1})$ given $\{\Delta_t = i, \Delta_{t-k} = j\}$ and $\{\Delta_{t-k} = j\}$ respectively. For all $\theta \in \Theta$, the densities $g_\theta(\epsilon_{t-1}, \ldots, \epsilon_{t-k-q+1}|\Delta_t = i, \Delta_{t-k} = j)$ and $g_\theta(\epsilon_{t-1}, \ldots, \epsilon_{t-k-q+1}|\Delta_{t-k} = j)$ are almost surely strictly positive. Moreover $(\epsilon_t, \ldots, \epsilon_{t-q+1}, \Delta_t)'$ is a Markovian process. Therefore, for all $i \in \mathscr{E}$, we have

$$P_\theta(\Delta_t = i|\epsilon_{t-1}, \epsilon_{t-2}, \ldots)$$

$$= \sum_{j=1}^{d} P_\theta(\Delta_t = i|\epsilon_{t-1}, \ldots, \epsilon_{t-k-q+1}, \Delta_{t-k} = j)P_\theta(\Delta_{t-k} = j|\epsilon_{t-1}, \epsilon_{t-2}, \ldots)$$

$$\geqslant \min_{j\in\mathscr{E}} P_\theta(\Delta_t = i|\epsilon_{t-1}, \ldots \epsilon_{t-k-q+1}, \Delta_{t-k} = j)$$

$$= \min_{j\in\mathscr{E}} \frac{g_\theta(\epsilon_{t-1}, \ldots, \epsilon_{t-k-q+1}|\Delta_t = i, \Delta_{t-k} = j)P_\theta(\Delta_t = i|\Delta_{t-k} = j)}{g_\theta(\epsilon_{t-1}, \ldots, \epsilon_{t-k-q+1}|\Delta_{t-k} = j)} > 0.$$

■

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LEMMA 2. *Let $\tilde{L}_\theta(\cdot)$ be the density of $(\epsilon_1, \ldots, \epsilon_n)$ given the σ-field generated by $\epsilon_0, \epsilon_{-1}, \ldots$. For all $\theta \in \Theta$,*

$$\lim_{n\to\infty} \frac{1}{n} \log \frac{\tilde{L}_\theta(\epsilon_1, \ldots, \epsilon_n)}{L_\theta(\epsilon_1, \ldots, \epsilon_n)} = 0 \qquad P_{\theta_0} - \text{a.s.}$$

PROOF. With the matrix norm used in the proof of Theorem 2, we have in view of (8)

$$\begin{aligned} &\min_j \pi(j) f_j(\epsilon_{1-q}, \ldots, \epsilon_1) \left\| \prod_{t=2}^{n} M_\theta(\epsilon_{n+2-t-q}, \ldots, \epsilon_{n-t+2}) \right\| \\ &\quad \leqslant L_\theta(\epsilon_1, \ldots, \epsilon_n) \\ &\quad \leqslant \max_j \pi(j) f_j(\epsilon_{1-q}, \ldots, \epsilon_1) \left\| \prod_{t=2}^{n} M_\theta(\epsilon_{n+2-t-q}, \ldots, \epsilon_{n-t+2}) \right\|. \end{aligned}$$

Therefore, when existing, we have

$$\lim_{n\to\infty} \frac{1}{n} \log L_\theta(\epsilon_1, \ldots, \epsilon_n) = \lim_{n\to\infty} \frac{1}{n} \log \left\| \prod_{t=2}^{n} M_\theta(\epsilon_{n+2-t-q}, \ldots, \epsilon_{n-t+2}) \right\|.$$

From Lemma 1 the same result holds for $\tilde{L}_\theta(\epsilon_1, \ldots, \epsilon_n)$:

$$\begin{aligned} &\lim_{n\to\infty} \frac{1}{n} \log \left\| \prod_{t=2}^{n} M_\theta(\epsilon_{n+2-t-q}, \ldots, \epsilon_{n-t+2}) \right\| \\ &\qquad = \lim_{n\to\infty} \frac{1}{n} \log \tilde{L}_\theta(\epsilon_1, \ldots, \epsilon_n) \\ &\qquad = E_{\theta_0} \log g_\theta(\epsilon_t | \epsilon_{t-1}, \epsilon_{t-2}, \ldots) \end{aligned}$$

where $g_\theta(\cdot|\epsilon_{t-1}, \epsilon_{t-2}, \ldots)$ denotes the density of ϵ_t given the σ-field generated by $\epsilon_{t-1}, \epsilon_{t-2}, \ldots$ (it is easy to show that $E_{\theta_0}|\log g_\theta(\epsilon_t|\epsilon_{t-1}, \epsilon_{t-2}, \ldots)| < \infty$). The conclusion follows. ∎

We are now in a position to show that the likelihood evaluated at θ is asymptotically strictly less than the likelihood evaluated at the true parameter value θ_0, when $\theta \neq \theta_0$. Write

$$O_n(\theta) = \frac{1}{n} \log \frac{L_\theta(\epsilon_1, \ldots, \epsilon_n)}{L_{\theta_0}(\epsilon_1, \ldots, \epsilon_n)}.$$

LEMMA 3. *For all $\theta \in \Theta$, almost surely*

$$\lim_{n\to\infty} O_n(\theta) \leqslant 0$$

and the limit is almost surely equal to zero if and only if $\theta = \theta_0$.

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PROOF. By Lemma 2, the ergodic theorem and the Jensen inequality we obtain

$$\lim_{n\to\infty} O_n(\theta) = E_{\theta_0} \log \frac{g_\theta(\epsilon_t|\epsilon_{t-1}, \epsilon_{t-2}, \ldots)}{g_{\theta_0}(\epsilon_t|\epsilon_{t-1}, \epsilon_{t-2}, \ldots)}$$

$$\leqslant \log E_{\theta_0} \frac{g_\theta(\epsilon_t|\epsilon_{t-1}, \epsilon_{t-2}, \ldots)}{g_{\theta_0}(\epsilon_t|\epsilon_{t-1}, \epsilon_{t-2}, \ldots)} = 0$$

with equality if and only if $\theta = \theta_0$. ∎

LEMMA 4. *For any $\theta_1 \in \Theta$, $\theta_1 \neq \theta_0$, there exists a neighborhood $V(\theta_1)$ of θ_1 such that*

$$\limsup_{n\to\infty} \sup_{\theta\in V(\theta_1)} O_n(\theta) < 0 \text{ a.s.}$$

PROOF. Let $V_r(\theta_1)$ be the open sphere with center θ_1 and radius $1/r$. Let

$$S_n^r(\epsilon_{1-q}, \ldots, \epsilon_n) = \sup_{\theta\in V_r(\theta_1)} \left\| \prod_{t=1}^{n} M_\theta(\epsilon_{n+1-q-t}, \ldots, \epsilon_{n+1-t}) \right\|.$$

As the matrix norm is multiplicative, we have

$$\log S_{n+k}^r(\epsilon_{1-q}, \ldots, \epsilon_{n+k}) \leqslant \log S_n^r(\epsilon_{1-q}, \ldots, \epsilon_n) + \log S_k^r(\epsilon_{n+1-q}, \ldots, \epsilon_{n+k})$$

for any positive integers n and r. From the Kingman ergodic theorem (1973) for subadditive processes we have almost surely

$$\lim_{n\to\infty} \frac{1}{n} \log S_n^r(\epsilon_{1-q}, \ldots, \epsilon_n) = \lambda_r(\theta_1) := \inf_{n\geqslant 1} \frac{1}{n} E_{\theta_0} \log S_n^r(\epsilon_{1-q}, \ldots, \epsilon_n).$$

Recall that almost surely

$$\lambda(\theta_0) := \lim_{n\to\infty} \frac{1}{n} \log \left\| \prod_{t=1}^{n} M_{\theta_0}(\epsilon_{n+1-q-t}, \ldots, \epsilon_{n+1-t}) \right\|$$

$$= \inf_{n\geqslant 1} \frac{1}{n} E_{\theta_0} \log \left\| \prod_{t=1}^{n} M_{\theta_0}(\epsilon_{n+1-q-t}, \ldots, \epsilon_{n+1-t}) \right\|.$$

Therefore, from Lemmas 2 and 3, there exist $\varepsilon > 0$ and $n_\varepsilon \in \mathbb{N}$ such that

$$\frac{1}{n_\varepsilon} E_{\theta_0} \log \left\| \prod_{i=1}^{n_\varepsilon} M_{\theta_1}(\epsilon_{n+1-q-t}, \ldots, \epsilon_{n+1-t}) \right\| < \lambda(\theta_0) - \varepsilon.$$

The dominated convergence theorem shows that, for r large enough,

$$\lambda_r(\theta_1) \leqslant \frac{1}{n_\varepsilon} E_{\theta_0} \log S_{n_\varepsilon}^r(\epsilon_{1-q}, \ldots, \epsilon_{n_\epsilon}) < \lambda(\theta_0) - \frac{\varepsilon}{2}.$$

∎

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Lemma 4 proves the strong consistency of the maximum likelihood estimator over any compact subset containing θ_0.

THEOREM 3. *Let Θ^* be a compact subset of Θ which contains θ_0 and let $(\hat{\theta}_n)$ be a sequence satisfying almost surely*

$$L_{\hat{\theta}_n}(\epsilon_1, \ldots, \epsilon_n) = \sup_{\theta \in \Theta^*} L_\theta(\epsilon_1, \ldots, \epsilon_n) \qquad \forall n.$$

Then $(\hat{\theta}_n)$ tends almost surely to θ_0 as $n \to \infty$.

5. PENALIZED MAXIMUM LIKELIHOOD ESTIMATOR

In this section a penalized likelihood estimator is proposed for identifying the orders of model (1). In order to obtain a computable and mathematically tractable likelihood function, we maintain the assumption of Section 4 that $p = 0$. The orders to be identified are therefore q and d. As the selected orders q and d are no longer supposed to be necessarily the true ones q_0 and d_0, we will write $\Theta(q, d)$, $\theta(q, d)$ and $\hat{\theta}_n(q, d)$ instead of Θ, θ and $\hat{\theta}_n$. It is supposed that all the previous assumptions still hold: the true parameter value $\theta_0(q_0, d_0)$ belongs to $\Theta(q_0, d_0)$, for all $\theta(q_0, d_0) \in \Theta(q_0, d_0)$; the associated Markov chain is irreducible and aperiodic; the ARCH parameters satisfy $\rho(\boldsymbol{P}) < 1$; if $g_{1,\theta(q_0,d_0)}(\epsilon_t|\epsilon_{t-1}, \epsilon_{t-2}, \ldots) = g_{1,\theta_0(q_0,d_0)}(\epsilon_t|\epsilon_{t-1}, \epsilon_{t-2}, , \ldots)$ $P_{\theta_0(q_0,d_0)}$ − a.s., then $\theta(q_0, d_0) = \theta_0(q_0, d_0)$. It is supposed that, for all $\theta(q, d) \in \Theta(q, d)$, the associated Markov chain is irreducible and aperiodic (when $d > 1$).

The maximum likelihood estimates are taken over compact subsets $\Theta^*(q, d)$ of the $\Theta(q, d)$s. It is supposed that $\theta_0(q_0, d_0)$ belongs to $\Theta^*(q_0, d_0)$. The possible orders q and d are supposed to be bounded above by some known orders Q and D. Following the general theory of model selection, consider the estimates $\hat{q}_n$, $\hat{d}_n$ which maximize

$$\log L_{\hat{\theta}_n(q,d)}(\epsilon_1, \ldots, \epsilon_n) - a_{q,d,n}$$

over $0 \leqslant q \leqslant Q$ and $1 \leqslant d \leqslant D$, where $a_{q,d,n}$ is a (possibly random) penalty term. It is supposed that $Q \geqslant q_0$ and $D \geqslant d_0$. Write $(q, d) \geqslant (q', d')$ if $q \geqslant q'$ and $d \geqslant d'$, and $(q, d) \not\geqslant (q', d')$ if $q < q'$ or $d < d'$. It is natural to choose a penalty term satisfying almost surely

$$a_{q,d,n} \geqslant 0 \qquad \text{and} \qquad a_{q,d,n} \geqslant a_{q',d',n} \qquad \text{when } (q, d) \geqslant (q', d'). \tag{11}$$

However, (11) will not be necessary. It will be only required that

$$\frac{a_{q,d,n}}{n} \to 0 \text{ a.s.} \qquad \text{as } n \to \infty. \tag{12}$$

The two most common choices for $a_{q,d,n}$ are $a_{q,d,n} = d^2 + qd$, (AIC, Akaike, 1969) and $a_{q,d,n} = (1/2)(\log n)(d^2 + qd)$, (BIC, Schwarz, 1978). It is also supposed that the $\Theta(q, d)$s are such that if

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$$(q, d) \not\geqslant (q_0, d_0)$$

then

$$\Theta(q, d) \text{ does not contain an element which is equivalent to } \theta_0(q_0, d_0) \quad (13)$$

in the sense that $\forall \theta(q, d) \in \Theta(q, d)$, $g_{1,\theta(q,d)}(\epsilon_t|\epsilon_{t-1}, \epsilon_{t-2}, \ldots)$ is not $P_{\theta_0(q_0,d_0)}$ – a.s. equal to $g_{1,\theta_0(q_0,d_0)}(\epsilon_t|\epsilon_{t-1}, \epsilon_{t-2}, \ldots)$. The assumption is an extension of Condition A. For instance, by the arguments given for Example 1, it can be shown that (13) holds in the following situation.

EXAMPLE 2. Let

$$\Theta(q, d) = \{\theta = (p(1, 2), \ldots, p(d, d), \omega', \alpha(1)', \ldots, \alpha(d)')'$$

$$\in]0, 1[^{d(d-1)} \times]0, +\infty[^{d} \times [0, +\infty[^{qd}:$$

$$\rho(\boldsymbol{P}) < 1, \omega(1) \leqslant \omega(2) \leqslant \cdots \leqslant \omega(d)\}.$$

Suppose that $\theta_0(q_0, d_0) \in \Theta(q_0, d_0)$ and that $\alpha_0(1) > 0, \ldots, \alpha_0(d_0) > 0$, $\omega_0(1) < \omega_0(2) < \cdots < \omega(d_0)$. Then (13) holds.

The following theorem shows that asymptotically $\hat{q}_n$ and $\hat{d}_n$ do not underestimate the true orders q_0 and d_0. This result extends that obtained by Leroux (1992a, 1992b) for mixing distributions and that obtained by Rydén (1994, 1995) for hidden Markov models.

THEOREM 4. *Under the previous assumptions on* (ϵ_t) *and on the parameter spaces* $\Theta(q, d)$ *and* $\Theta^*(q_0, d_0)$, *we have*

$$\liminf_{n\to\infty} \hat{q}_n \geqslant q_0 \qquad \textit{and} \qquad \liminf_{n\to\infty} \hat{d}_n \geqslant d_0 \text{ a.s.}$$

whenever (12) holds.

Consider the Kullback-Leibler distance between the true parameter $\theta_0 = \theta_0(q_0, d_0)$ and another parameter $\theta(q, d)$

$$K\{\theta_0, \theta(q, d)\} = -E_{\theta_0} \log \frac{g_{1,\theta(q,d)}}{g_{1,\theta_0}}(\epsilon_t|\epsilon_{t-1}, \epsilon_{t-2}, , \ldots).$$

Write also

$$K\{\theta_0, \Theta^*(q, d)\} = \inf_{\theta(q,d)\in\Theta^*(q,d)} K\{\theta_0, \theta(q, d)\}.$$

Using the arguments given by Leroux (1992b, Lemma 3) and Rydén (1995, Lemma 2) we show that there exists a $\theta_0(q, d) \in \Theta^*(q, d)$ such that $K\{\theta_0, \theta_0(q, d)\} = K\{\theta_0, \Theta^*(q, d)\}$. For each (q, d), write $\Theta_0(q, d)$ as the set of these $\theta_0(q, d)$. Extending the results of Section 4 we give the following lemma.

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LEMMA 5. *Under the assumptions of Theorem 4 we have*

$$\lim_{n\to\infty} \frac{1}{n} \log L_{\hat{\theta}_n(q,d)}(\epsilon_1, \ldots, \epsilon_n) = E_{\theta_0} \log g_{1,\theta_0(q,d)}(\epsilon_t|\epsilon_{t-1}, \epsilon_{t-2}, , \ldots) \text{ a.s.}$$

PROOF. Using the arguments given in the proof of Lemma 4, it can be shown that for all $\theta_0(q, d) \in \Theta_0(q, d)$ and all $\theta_1(q, d) \in \Theta^*(q, d)$, $\theta_1(q, d) \notin \Theta_0(q, d)$, there exists a neighborhood $V(\theta_1)$ of $\theta_1(q, d)$ such that

$$\limsup_{n\to\infty} \sup_{\theta(q,d)\in V(\theta_1)} \left\{ \frac{1}{n} \log L_{\theta(q,d)}(\epsilon_1, \ldots, \epsilon_n) - \frac{1}{n} \log L_{\theta_0(q,d)}(\epsilon_1, \ldots, \epsilon_n) \right\} < 0 \text{ a.s.} \tag{14}$$

Let $V_r(\theta)$ be the open sphere with center θ and radius $1/r$. By a usual compactness argument, there exists a finite subset of $\Theta_0(q, d)$, $\{\theta_0^{(1)}(q, d), \ldots, \theta_0^{(k)}(q, d)\}$, such that $\Theta_0(q, d)$ is covered by $\mathscr{S} = \{V_r[\theta_0^{(1)}(q, d)], \ldots, V_r(\theta_0^{(k)}(q, d))\}$. By the same argument, $\Theta^*(q, d)$ is covered by $\mathscr{S}$ and a finite subset of the $V(\theta_1(q, d))$, $\theta_1(q, d) \in \Theta^*(q, d)\backslash\mathscr{S}$, satisfying (14) for all $\theta_0(q, d) = \theta_0^{(i)}(q, d)$. We deduce that

$$\limsup_{n\to\infty} \sup_{\theta(q,d)\in\Theta^*(q,d)} \frac{1}{n} \log L_{\theta(q,d)}(\epsilon_1, \ldots, \epsilon_n) \leqslant \max_{i=1,\ldots,k} E_{\theta_0} \sup_{\theta\in V_r\{\theta_0^{(i)}(q,d)\}} \log g_{1,\theta}(\epsilon_t|\epsilon_{t-1}, \epsilon_{t-2}, \ldots).$$

By the dominated convergence theorem we have

$$\lim_{r\to\infty} E_{\theta_0} \sup_{\theta\in V_r\{\theta_0^{(i)}(q,d)\}} \log g_{1,\theta}(\epsilon_t|\epsilon_{t-1}, \epsilon_{t-2}, \ldots) = E_{\theta_0} \log g_{1,\theta_0^{(1)}(q,d)}(\epsilon_t|\epsilon_{t-1}, \epsilon_{t-2}, \ldots).$$

The conclusion follows. ∎

PROOF OF THEOREM 4. The Jensen inequality and the identifiability condition (13) show that $K\{\theta_0, \theta(q, d)\} \geqslant 0$ with strict inequality when $(q, d) \not\geqslant (q_0, d_0)$. Therefore Lemma 5 and (12) entail that, for each $(q, d) \not\geqslant (q_0, d_0)$, we have almost surely

$$\log L_{\hat{\theta}_n(q,d)}(\epsilon_1, \ldots, \epsilon_n) - a_{q,d,n} < \log L_{\hat{\theta}_n(q_0,d_0)}(\epsilon_1, \ldots, \epsilon_n) - a_{q_0,d_0,n}$$

for large n, since

$$\lim_{n\to\infty} \frac{1}{n} \left\{ \log \frac{L_{\hat{\theta}_n(q,d)}(\epsilon_1, \ldots, \epsilon_n)}{L_{\hat{\theta}_n(q_0,d_0)}(\epsilon_1, \ldots, \epsilon_n)} - a_{q,d,n} + a_{q_0,d_0 n} \right\} = -K\{\theta_0, \theta(q, d)\} < 0 \text{ a.s.}$$

The conclusion follows since the set of the (q, d)s which are not greater than (q_0, d_0) is finite (it contains $(Q + 1)D - (Q - q_0 + 1)(D - d_0 + 1)$ elements).

∎

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The same arguments show that the estimated order are asymptotically the true ones when (13) also holds for $(q, d) \geqslant (q_0, d_0)$:

if

$$(q, d) \neq (q_0, d_0) \tag{15}$$

then

$$\Theta(q, d) \text{ does not contain an element equivalent to } \theta_0(q_0, d_0).$$

Of course (15) is not true in the case of Example 2, since, for instance, an element of $\Theta(q_0, d_0 + 1)$, equivalent to $\theta_0(q_0, d_0)$, can be obtained by choosing $\omega(d_0 + 1) = \omega_0(d_0)$, $\alpha(d_0 + 1) = \alpha(d_0)$, and obvious values for the other parameters.

EXAMPLE 3. Let

$$\Theta(q, d) = \{\theta = (p(1, 2), \ldots, p(d, d), \omega', \alpha(1)' \ldots, \alpha(d)')'$$
$$\in]0, 1[^{d(d-1)} \times]0, +\infty[^{d} \times]0, +\infty[^{qd}:$$
$$\rho(\boldsymbol{P}) < 1, \omega(1) < \omega(2) < \cdots < \omega(d)\}.$$

Suppose that $\theta_0(q_0, d_0) \in \Theta(q_0, d_0)$. Then (15) holds.

Using the proof of Theorem 4, the next result follows.

THEOREM 5. *If, with the assumptions of Theorem 4, (13) is replaced by (15), then*

$$\lim_{n\to\infty} \hat{q}_n = q_0 \qquad \textit{and} \qquad \lim_{n\to\infty} \hat{d}_n = d_0 \text{ a.s.}$$

6. NUMERICAL ILLUSTRATIONS

We propose, in this section, some numerical illustrations in order to demonstrate that the proposed procedures are tractable (at least for a moderate number of regimes, say $d \leqslant 3$) and to show that model (1) can capture some special features of financial time series. Three Monte Carlo experiments were conducted. The first one illustrates the performance of the maximum likelihood estimator in finite samples. The second shows how examination of the maximum likelihood estimates can reveal a misspecification of the model. The third is a comparison of AIC and BIC. Finally, our model is fitted to a French financial time series.

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6.1. *Simulation experiments*

First, 100 independent trajectories of size 1000 of model (1) were simulated, with $d = 2$, $q = 1$, $p = 0$, $p(1, 2) = 0.01$, $p(2, 2) = 0.99$, $\omega(1) = 1$, $\omega(2) = 100$, $\alpha_1(1) = 0.6$ and $\alpha_1(2) = 0$. The simulations and the maximum likelihood were obtained by means of the NAG Fortran workstation library (routine G05DDF for the Gaussian random variables generator and E04UCF for the log-likelihood maximization). The linear constraints $0 \leqslant p(i, j) \leqslant 1$, $\sum_{j=2}^{d} p(i, j) \leqslant 1$, $\omega(i) \geqslant 0.01$, $\alpha_j(i) \geqslant 0$, $\sum_{j=1}^{q} \alpha_j(i) \leqslant 1$ and $\omega(1) \leqslant \omega(2) \leqslant \cdots \leqslant \omega(d)$ were imposed. Several starting values were used to circumvent the eventuality of a local maximum in the optimization step. For each parameter, Figure 1 displays the boxplot of the 100 differences between the true value of the parameter and its estimate. As expected the estimation errors move around zero.

In the second experiment we fitted various misspecified models of the form (1) to one of the previous simulations. Table I and other simulations (not reported here in the interests of brevity) show that when we fit a standard GARCH model (i.e. $d = 1$) to simulated data sets with at least two very different regimes, we typically obtain estimates close to the nonstationary region ($\sum_i \alpha_i + \beta_i \simeq 1$). Such parameter estimates have been observed for many financial time series by earlier researchers. Without the linear constraint $\sum_i \alpha_i + \beta_i \leqslant 1$, the parameter estimates are generally outside the stationary region. This leads us to think that GARCH parameter estimates close to the nonstationary region could indicate that an alternative model, such as (1), should be considered by the practitioner. It should also be noted that several GARCH parameter estimates are exactly zero. This is due to the constraints $\alpha_j(i) \geqslant 0$. Another interesting output is that, when $d = 3$, the second and third estimated GARCH regimes are exactly the same. This clearly shows that $d = 3$

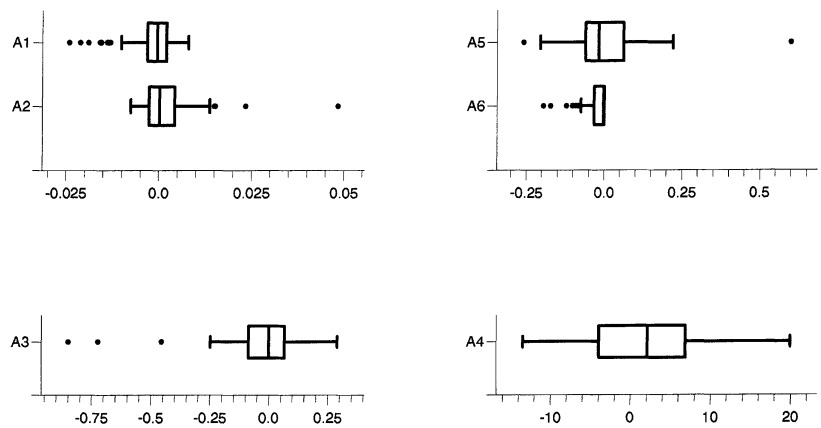


FIGURE 1. Maximum likelihood estimates of 100 independent trajectories of size 1000 of model (1) (with $d = 2$, $q = 1$ and $p = 0$). A1, A2, A3, A4, A5 and A6 correspond respectively to the boxplots of the 100 estimation errors of parameters $p(1, 2)$, $p(2, 2)$, $\omega(1)$, $\omega(2)$, $\alpha_1(1)$ and $\alpha_1(2)$.

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TABLE I

MAXIMUM LIKELIHOOD ESTIMATES OF VARIOUS MODELS FOR ONE TRAJECTORY OF SIZE 1000 OF MODEL (1) (WITH $d = 2$, $q = 1$ AND $p = 0$)

Model	Parameter estimates
$d = 1, q = 1, p = 1$	$\omega = 0.83 \quad \alpha(1) = 0.312 \quad \beta(1) = 0.688$
$d = 1, q = 2, p = 0$	$\omega = 9.22 \quad \alpha(1) = (0.482, 0.518)$
$d = 2, q = 2, p = 0$	$\omega = (0.99, 96.25)$
	$\alpha(1) = (0.615, 0.022) \quad \alpha(2) = (0.0, 0.009)$
	$p(1, 1) = 0.986 \quad p(1, 2) = 0.014 \quad p(2, 1) = 0.014 \quad p(2.2) = 0.986$
$d = 3, q = 1, p = 0$	$\omega = (1.02, 96.91, 96.91)$
	$\alpha(1) = 0.623 \quad \alpha(2) = 0.0 \quad \alpha(3) = 0.0$
	$p(1, 1) = 0.986 \quad p(1, 2) = 0.014 \quad p(1, 3) = 0.0$
	$p(2, 1) = 0.0 \quad p(2, 2) = 0.945 \quad p(2, 3) = 0.055$
	$p(3, 1) = 0.086 \quad p(3, 2) = 0.209 \quad p(3, 3) = 0.705$

regimes is not more plausible than $d = 2$. But it also entails that an overparameterization ($d = 3$ instead of $d = 2$) does not affect the forecasting performance of the model. This amazing and nice feature of the maximum likelihood estimates has already been noted by Francq and Roussignol (1997) in the framework of hidden Markov models.

The third experiment consists in fitting 12 models of the form (1) (with $d = 1, 2, 3$ and $q = 0, 1, 2, 3$) for the 100 generated data sets of the first experiment (and for subsets of size $n = 100$). It is not possible to draw definitive conclusions from this single experiment but AIC and BIC seem both to have quite good behavior (Figure 2). Theorem 4 shows that they do not

(a)

d \ q	0	1	2	3
1	13	12	2	2
2	26	39	3	1
3	2	0	0	0

(b)

d \ q	0	1	2	3
1	20	12	5	2
2	28	31	0	0
3	2	0	0	0

(c)

d \ q	0	1	2	3
1	0	0	0	0
2	1	92	7	0
3	0	0	0	0

(d)

d \ q	0	1	2	3
1	0	0	0	0
2	2	97	1	0
3	0	0	0	0

FIGURE 2. Comparison between AIC and BIC for 100 independent trajectories of size $n = 100$ and $n = 1000$ of model (1) (with $d = 2$, $q = 1$ and $p = 0$). Among the 100, the number of selected models with orders d and q is at the intersection of the dth row and the qth column: (a), $n = 100$, AIC; (b) $n = 100$, BIC; (c) $n = 1000$, AIC; (d) $n = 1000$, BIC.

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asymptotically underestimate the orders and, in the framework of this experiment, it is worth noting that the criteria are not visibly inclined to automatically overestimate the orders.

6.2. *French CAC 40 time series*

The French CAC 40 is a daily stock index computed from 40 stocks traded on the montly settlement market. It is built to represent the whole range of equities listed on the Paris Bourse and it is used in futures and options markets. Our data, denoted by p_t, $t = 1, \ldots, 1286$, cover the period from 1 June 1988 (date of inauguration of the index) to 31 December 1993. Let $\epsilon_t = 100(\ln p_t - \ln p_{t-1})$ denote the compounded return at time t.

In Table II, various fitted models of the form (1) are reported. First a standard GARCH(1, 1) ($d = 1$) is estimated under the second-order stationarity condition ($\alpha(1) + \beta(1) < 1$). In accordance with many empirical studies on similar data, the estimates are very close to the stationary region indicating strong persistence in variance. Other results not reported here on serveral ARCH(q) confirm this evidence.

Allowing for $d = 2$ regimes in the absence of GARCH($p = q = 0$) leads to important differences in the estimates of the noise variances $\omega(1)$ and $\omega(2)$. Concerning the transition probabilities, it is worth noting that the probability of staying in any regime is close to 1, which confirms the persistence property. Introducing an ARCH(1) effect in each regime ($d = 2$, $p = 0$, $q = 1$) does not alter these remarks: the two regimes remain very different and persistent. Moreover a very small conditional heteroskedasticity within each regime is observed. Increasing the ARCH order seems to be ineffective: the estimated models are indeed the same for $q = 1$ and $q = 2$.

The other models presented in Table II are obtained by increasing the number of regimes. The presence of five regimes seems to be rejected by the data: for $d = 5$, $q = p = 0$, the marginal variances in the last two regimes are equal ($\omega(4) = \omega(5)$). The results obtained for $d = 3$ and $d = 4$ indicate that the inclusion of four regimes is worth consideration. First, the estimated marginal variances ($\omega(i)/\{1 - \alpha(i)\}$, $i = 1, \ldots, 4$) are very different. Moreover, from the transition matrix, the first three regimes are extremely persistent. In contrast, only a few successive observations are likely to belong to the more volatile regime. The third regime is charaterized by a coefficient $\alpha(1)$ not close to zero, which justifies taking conditional heteroskedasticity into account.

Figure 3 displays the series ϵ_t, $t = 2, \ldots, 1286$ (full line). In order to identify the regime changes, using dotted lines we show ± 2 times the conditional standard deviation (within each regime) of the estimated ARCH model for the regime that maximizes the smoothed probabilities (computed at each date using all the observations). The graph confirms the above-mentioned remarks: little persistence and little occurrence of the fourth regime; presence of ARCH in the third one; strong occurrence of the first two regimes with no ARCH effect.

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TABLE II

MAXIMUM LIKELIHOOD ESTIMATES OF VARIOUS MODELS FOR THE FRENCH CAC 40 TIME SERIES

Model	Parameter estimates
$d = 1,\ q = 1,\ p = 1$	$\omega = 0.13\ \ \alpha(1) = 0.184\ \ \beta(1) = 0.739$
$d = 2,\ q = 0,\ p = 0$	$\omega = (0.81, 4.46)$ $p(1, 1) = 0.977\ \ p(1, 2) = 0.023\ \ p(2, 1) = 0.102\ \ p(2, 2) = 0.898$
$d = 2,\ q = 1,\ p = 0$	$\omega = (0.76, 3.71)$ $\alpha(1) = 0.05\ \ \alpha(2) = 0.10$ $p(1, 1) = 0.979\ \ p(1, 2) = 0.021\ \ p(2.1) = 0.084\ \ p(2, 2) = 0.916$
$d = 2,\ q = 2,\ p = 0$	$\omega = (0.76, 3.70)$ $\alpha(1) = (0.05, 0.0)\ \ \alpha(2) = (0.10, 0, 00)$ $p(1, 1) = 0.979\ \ p(1, 2) = 0.021\ \ p(2, 1) = 0.084\ \ p(2, 2) = 0.916$
$d = 3,\ q = 0,\ p = 0$	$\omega = (0.42, 1.27, 6.77)$ $p(1, 1) = 0.972\ \ p(1, 2) = 0.015\ \ p(1, 3) = 0.013$ $p(2, 1) = 0.008\ \ p(2, 2) = 0.982\ \ p(2, 3) = 0.010$ $p(3, 1) = 0.038\ \ p(3, 2) = 0.080\ \ p(3, 3) = 0.882$
$d = 3,\ q = 1,\ p = 0$	$\omega = (0.41, 1.25, 5.75)$ $\alpha(1) = 0.01\ \ \alpha(2) = 0.0\ \ \alpha(3) = 0.09$ $p(1, 1) = 0.973\ \ p(1, 2) = 0.014\ \ p(1, 3) = 0.013$ $p(2, 1) = 0.007\ \ p(2, 2) = 0.983\ \ p(2, 3) = 0.010$ $p(3, 1) = 0.037\ \ p(3, 2) = 0.071\ \ p(3, 3) = 0.892$
$d = 3,\ q = 2,\ p = 0$	$\omega = (0.41, 1.25, 5.75)$ $\alpha(1) = (0.01, 0.0)\ \ \alpha(2) = (0.0, 0.0)\ \ \alpha(3) = (0.09, 0.0)$ $p(1, 1) = 0.973\ \ p(1, 2) = 0.014\ \ p(1.3) = 0.013$ $p(2, 1) = 0.007\ \ p(2, 2) = 0.983\ \ p(2, 3) = 0.010$ $p(3, 1) = 0.037\ \ p(3, 2) = 0.071\ \ p(3, 3) = 0.892$
$d = 4,\ q = 1,\ p = 0$	$\omega = (0.40, 1.17, 2.90, 12.11)$ $\alpha(1) = 0.01\ \ \alpha(2) = 0.0\ \ \alpha(3) = 0.23\ \ \alpha(4) = 0.06$ $p(1, 1) = 0.972\ \ p(1, 2) = 0.016\ \ p(1, 3) = 0.009\ \ p(1, 4) = 0.003$ $p(2, 1) = 0.006\ \ p(2, 2) = 0.983\ \ p(2, 3) = 0.001\ \ p(2, 4) = 0.010$ $p(3, 1) = 0.0\ \ p(3, 2) = 0.028\ \ p$: $(3, 3) = 0.972\ \ p(3, 4) = 0.0$ $p(4, 1) = 0.178\ \ p(4, 2) = 0.139\ \ p(4, 3) = 0.0\ \ p(4, 4) = 0.682$
$d = 5,\ q = 0,\ p = 0$	$\omega = (0.44, 1.13, 2.73, 10.88, 10.88)$ $p(1, 1) = 0.974\ \ p(1, 2) = 0.015\ \ p(1, 3) = 0.0\ \ p(1, 4) = 0.0$ $p(1, 5) = 0.010$ $p(2, 1) = 0.006\ \ p(2, 2) = 0.980\ \ p(2, 3) = 0.004\ \ p(2, 4) = 0.010$ $p(2, 5) = 0.0$ $p(3, 1) = 0.0\ \ p(3, 2) = 0.021\ \ p(3, 3) = 0.964\ \ p(3, 4) = 0.015$ $p(3, 5) = 0.0$ $p(4, 1) = 0.159\ \ p(4, 2) = 0.441\ \ p(4, 3) = 0.0\ \ p(4, 4) = 0.400$ $p(4, 5) = 0.0$ $p(5, 1) = 0.040\ \ p(5, 2) = 0.0\ \ p(5, 3) = 0.079\ \ p(5, 4) = 0.0$ $p(5, 5) = 0.880$

7. DISCUSSION

In this paper, consistency of the maximum likelihood estimator was established in the case of ARCH regimes. Two important issues not considered here are the asymptotic distribution of the maximum likelihood estimator and the estimation of the parameters in the case of GARCH regimes. The former issue has been

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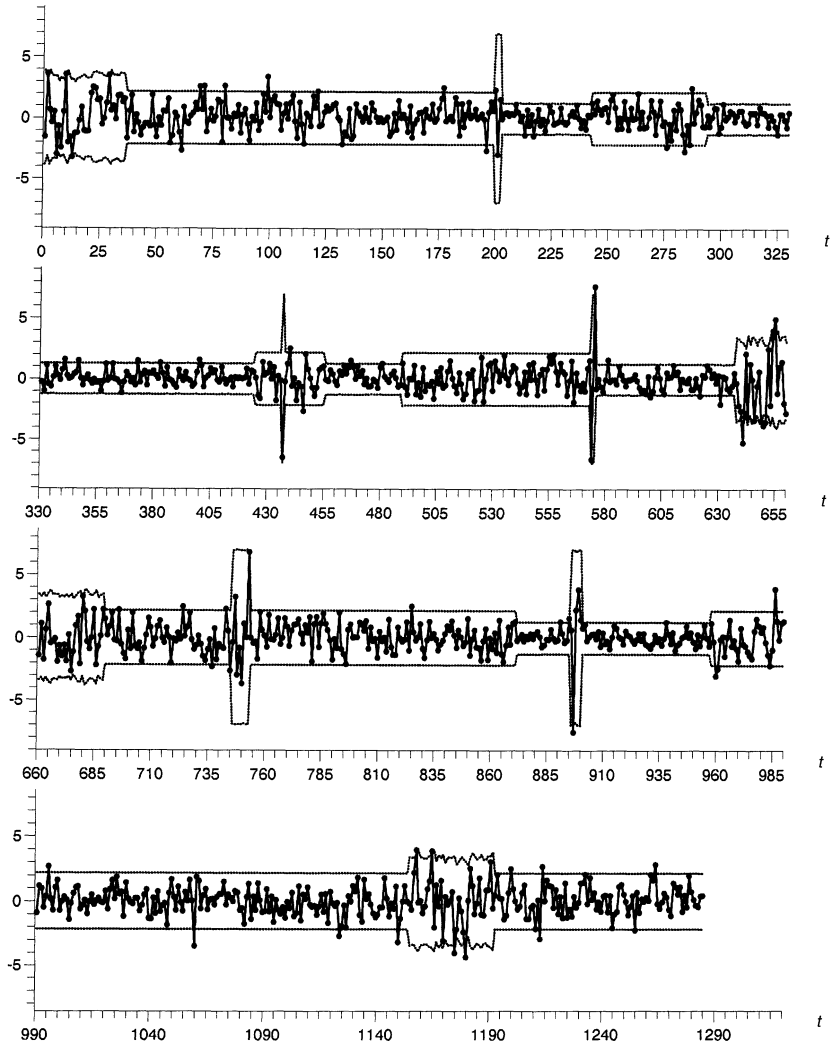


FIGURE 3. CAC 40 time series from 2 June 1988 to 31 December 1993. The dotted lines correspond to ± 2 times the conditional standard deviation of the estimated ARCH model for the regime that maximizes the smoothed probabilities.

addressed recently by Bickel *et al.* (1998) in the case of i.i.d. regimes (hidden Markov models). However, this approach cannot be straightforwardly adapted to our framework. Concerning the second issue, the maximum likelihood approach seems intractable in the case of GARCH regimes, because the regimes are non-Markovian. We are currently working on an estimation method based on linear

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representations of the process (see Francq and Zakoïan, 1999). This procedure requires an extension of the stationarity results of the present paper.

NOTE

1. These Markov-switching models clearly have many common features with the hidden Markov models introduced by Baum and Petrie (1966) and successfully used in the field of speech pattern modeling (see for instance Rabiner and Juang, 1986; Elliot *et al.*, 1995).

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